FlatHill

Robot hardware, built for scale.

A U.S. robotics hardware supply platform.
Proprietary joints. Manufacturing expertise.
An AI workflow from concept to production.

The bottleneck is repeatable hardware.

A working robot becomes a business when its cost, quality and supply can repeat across every production batch.

01	**Manufacturing expertise**	The team brings prior robot delivery experience into design decisions, supplier selection and factory processes.
02	**Owned module architecture**	Micro drives and larger joint families provide reusable building blocks across hands, arms and mobile platforms.
03	**A U.S. production path**	Selected global inputs feed documented U.S. manufacturing, supported by product-specific sourcing and authorization work.

ROSE turns each engineering and production cycle into reusable manufacturing knowledge.

A team shaped by robot delivery.

150+

robots delivered in prior career programs

Team-provided track record in patrol robotics.
Not FlatHill shipments.

Jiyuan Song 宋纪元

Robotics PhD, Xi’an Jiaotong University. Prior GM research experience. Serial robotics engineering and venture experience.

Yao Tu 屠尧

10 years of embedded systems, sensor fusion and motion planning experience, as described in team materials.

Hebing Liu 刘贺兵

Mobile robotics and autonomy experience. Team materials attribute 150+ patrol robot deliveries to prior programs.

The engineering bench spans mechanisms, motor control, embedded electronics and system integration.

Internal: original DexHand deck, team slide 3. Career experience is management-provided, not independently audited.

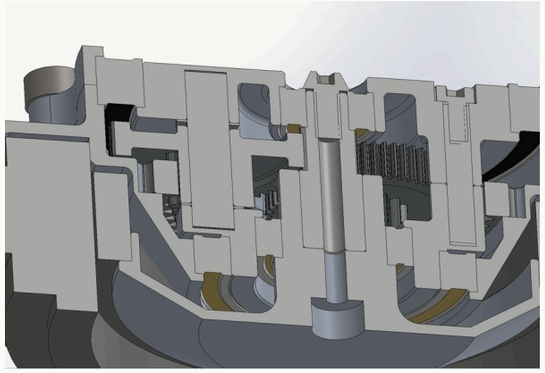
One actuation foundation. Two scales of motion.



MICRO DRIVE FAMILY

M01 / M01 Neo / M02

Compact joint architecture for fingers and small end effectors. Integrated transmission, sensing and control.



LARGE JOINT FAMILY

WZ53 / WZ79 / WZ119

Reusable joint architecture for larger mechanisms. Source design targets span 7 / 15 / 60 N·m.

Self-developed module designs allow FlatHill to control interfaces, tolerances, firmware and production changes.

Internal: joint module development PDF and DexHand deck. Large-joint figures are design targets. Production maturity varies by SKU.

Cost control starts before the factory.

A COMMON COST MODEL FOR EVERY RELEASED SKU

Landed parts + Build & test + Yield loss + Field support

Measure cost per good unit. Keep this operating-cost model separate from the origin-content calculation.

01	**Fewer unique parts**	Reuse actuator families, connectors and fixtures across products. Reduce tooling, inventory and redesign work.
02	**Design for the process**	Match tolerances to function. Standardize assembly sequence and end-of-line tests before scaling orders.
03	**Measure the delivered cost**	Track purchase cost, duties, yield loss, rework and warranty cost for each released configuration.

The economic goal: lower total delivered cost with repeatable quality.

Four platforms. Shared engineering.



01

Dexterous hands

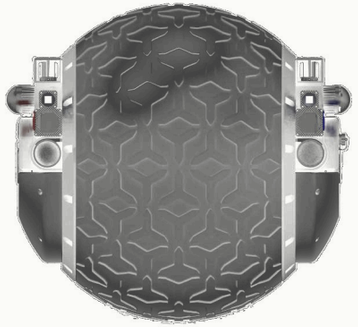
Micro drives and replaceable finger modules.



02

Quadrupeds

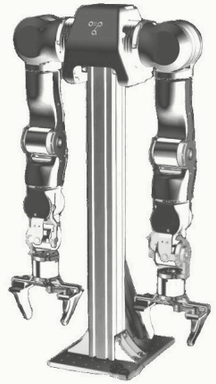
Joint assemblies for mobile inspection platforms.



03

Spherical robots

Enclosed mobility with drive and stabilization systems.



04

Dual-arm platforms

Manipulation systems with modular joints and end effectors.

Platform families demonstrate integration. Standard modules create the repeatable supply business.

Existing FlatHill product renders. Specifications, readiness and authorization are configuration-specific.

Start with components.
Expand with customer programs.

01	**First purchase: engineering teams**	Sell evaluation modules and hand / arm development kits to U.S. robotics OEMs, AI labs and integrators.
02	**Production adoption: repeat orders**	Convert validated designs into volume component supply, configured assemblies and paid engineering work.
03	**Expansion: software and capacity**	Add ROSE workflow subscriptions and manufacturing services as the software and factory processes mature.

NEAR-TERM REVENUE
Hardware margin + engineering fees

RECURRING OPPORTUNITY
Module replenishment + ROSE

U.S. manufacturing. A disciplined global input strategy.

PHASE 1 / NOW THROUGH 2027 / PLAN

01

Source selected inputs

Qualify commodity electronics and motor inputs, including eligible China-sourced parts. Record maker, origin and firmware provenance.

02

Manufacture in the U.S.

Build qualifying component assemblies and end products through U.S. operations or approved manufacturing partners.

03

Release a documented SKU

Tie supplier declarations and cost records to serial numbers, tests and the applicable authorization path.

>75% U.S.-supplier purchasing objective

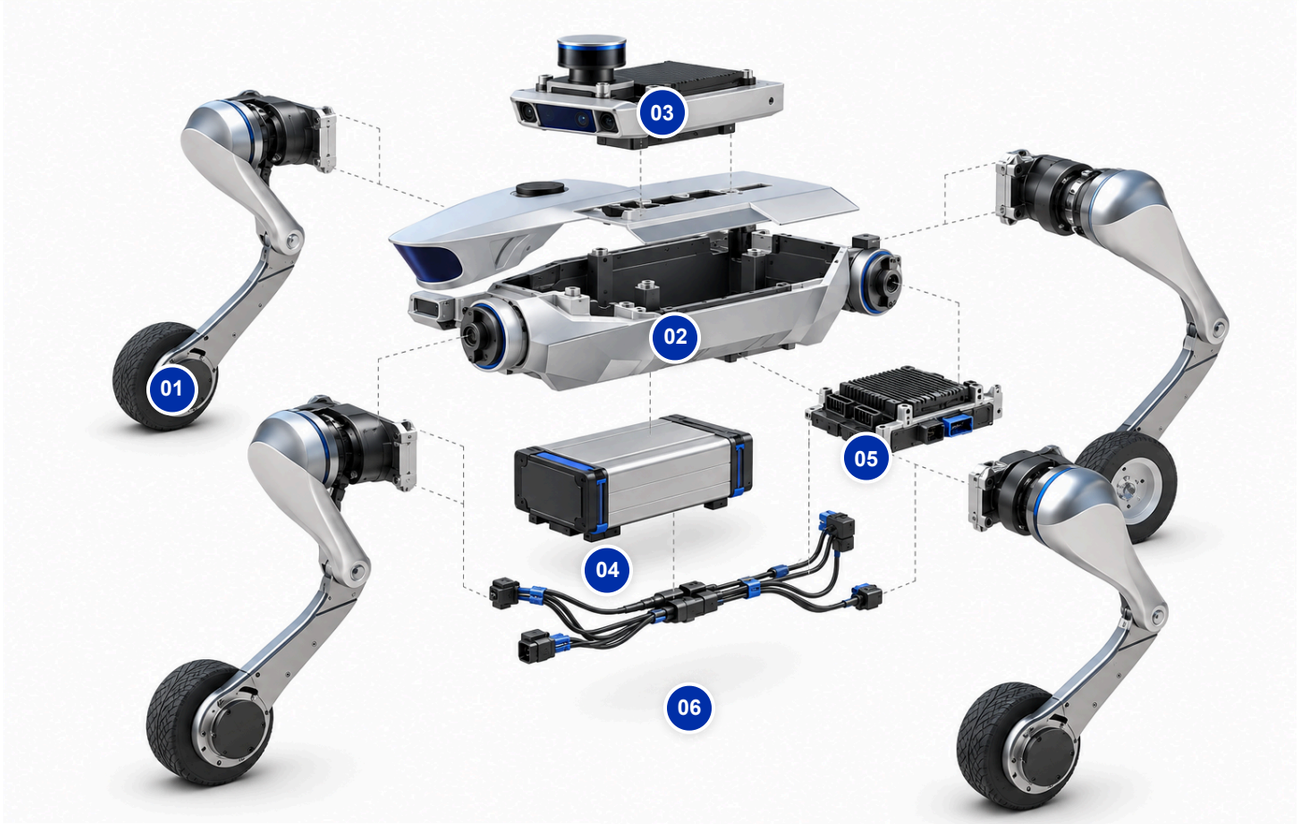
80% Illustrative U.S.-qualifying component-cost target

These are separate measures. A U.S. invoice alone does not establish U.S. manufacture or component origin. Actual sourcing remains to be validated.

FAR 25.101 and 25.003. See appendices for calculation, assumptions and product scope.

Quadruped: localize the highest-value assemblies.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Joint / leg assemblies	42%	42 pp
02 Chassis and structures	16%	16 pp
03 Compute and sensing	14%	10 pp
04 Battery assembly	12%	12 pp
05 Control / power boards	10%	0 pp
06 Harness and connectors	6%	0 pp
Total	**100%**	**80 pp**

*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

Joint assemblies are the largest dependency. Imported finished joints cannot become domestic merely through a U.S. reseller.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Dexterous hand: cost lives in the micro joints.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Micro joint modules	48%	48 pp
02 Finger / palm mechanics	22%	22 pp
03 Tactile / force sensing	12%	2 pp
04 Control electronics	10%	0 pp
05 Wrist and harness	8%	8 pp
Total	**100%**	**80 pp**

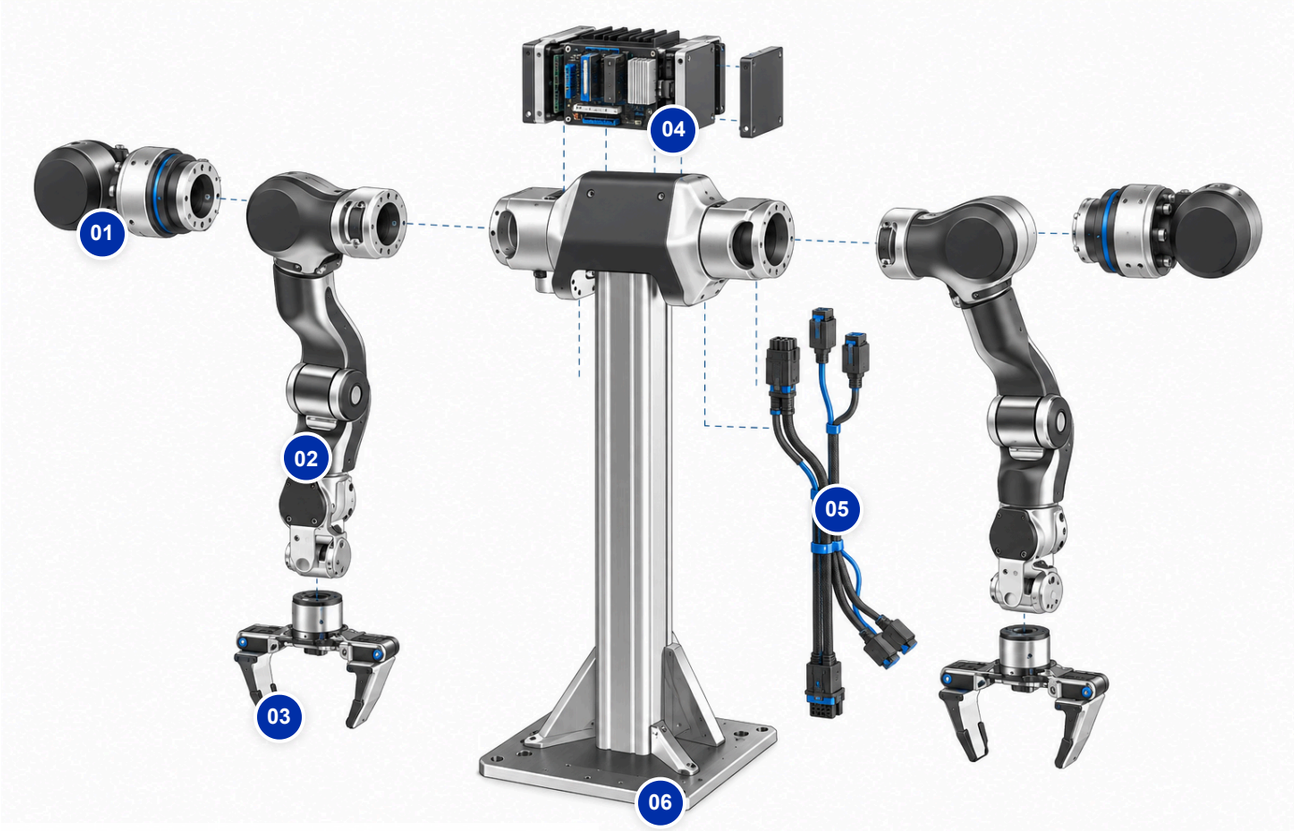
*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A shared micro-drive design concentrates cost engineering, calibration and replacement parts in one module family.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Dual arms: reuse joints, links and test fixtures.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Joint modules	44%	44 pp
02 Links and housings	18%	18 pp
03 End effectors	12%	12 pp
04 Control / compute	10%	0 pp
05 Power / harness	10%	0 pp
06 Pedestal and base	6%	6 pp
Total	**100%**	**80 pp**

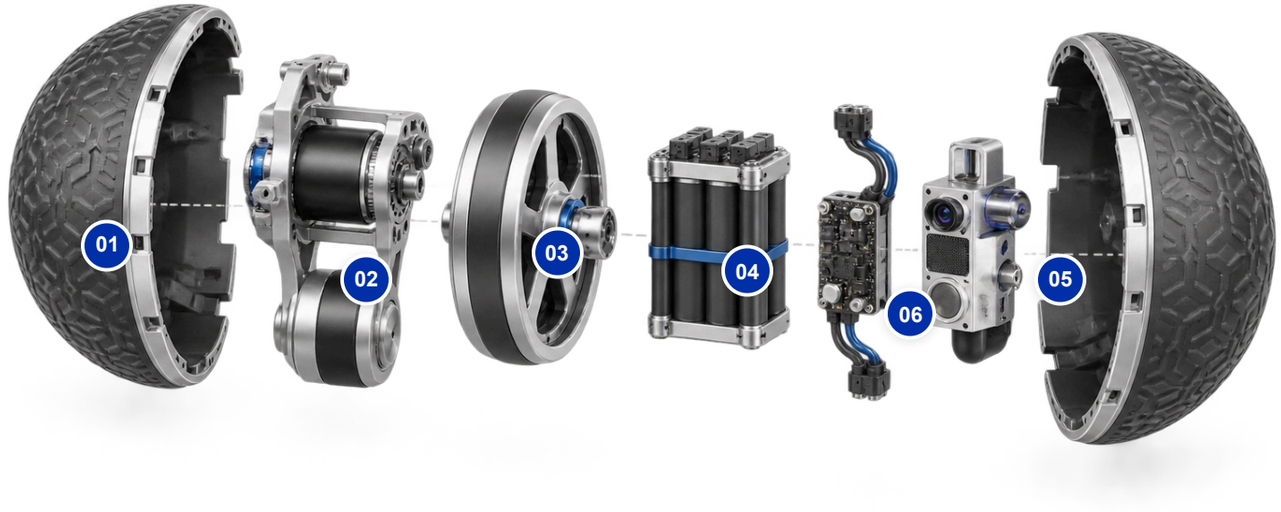
*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A fixed industrial arm configuration has a different FCC scope analysis from a mobile robot carrying the same arm.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Spherical robot: separate mobility from payload.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Shell and structure	24%	24 pp
02 Pendulum drive assembly	24%	24 pp
03 Stabilization / flywheel	14%	14 pp
04 Battery assembly	18%	18 pp
05 Compute / sensing	12%	0 pp
06 Power boards / harness	8%	0 pp
Total	**100%**	**80 pp**

*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A common drive core can support multiple sensing payloads. Origin and authorization follow the shipped configuration.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Compliance is a product release discipline.

80%

Illustrative domestic component-cost target.
Supported by actual U.S. manufacture and an evidence file for each SKU.

01	**Classify the finished product**	The current FCC determination targets defined mobile robotic devices. Fixed industrial robots are excluded from that definition.
02	**Validate manufacture and component cost**	The referenced FAR test generally requires U.S. manufacture and domestic component cost above 65% through 2028, then above 75% from 2029. Exceptions require review.
03	**Complete the authorization path**	Covered List treatment and normal equipment authorization remain separate release checks. Conditional approval is a product-specific alternative where available.

FCC DA 26-786, Appendix C · FAR 25.101 · FCC DA 26-996. Regulatory snapshot: 12 Oct 2026.

Design control makes localization possible.

FlatHill’s proposed advantage is a transferable engineering package: module designs, drawings, firmware and manufacturing know-how.

01	**Own the engineering decisions**	Management reports self-developed joint module families and complete technical drawings. Confirm assignments and third-party licenses per product.
02	**Build the U.S. rights package**	Maintain an inventor and employer chain of title, U.S. filing strategy and product-specific freedom-to-operate analysis.
03	**Keep capitalization transparent**	Management states a U.S.-first financing and commercialization strategy. Confirm the FlatHill entity’s cap table separately from team members’ prior ventures.

Investors can diligence the rights, the design revisions and the manufacturing transfer together.

Management representations, October 2026. Executed IP assignments, U.S. filing receipts and a counsel FTO opinion were not reviewed for this deck.

ROSE connects engineering to manufacturing.

PHASE 2 / COMMERCIALIZATION AFTER 2027 / ROADMAP

ROSE

DRAFTGPT WORKFLOW 01

From scratch to demo

Turn requirements into a buildable design and a tested prototype.

DRAFTGPT WORKFLOW 02

From demo to mass production

Turn a working design into a controlled manufacturing package.

Management reports an early software prototype. The workflows shown here define the intended product scope, not verified shipped functionality.

From a requirement to a testable prototype.

ROSE workflow 01 / From scratch to demo

01

Describe the job

Payload, reach, duty cycle, environment and target cost become explicit requirements.

02

Generate a build plan

Suggest module choices, mechanical interfaces, control code and a preliminary BOM.

03

Simulate and review

Check motion, loads, interfaces and failure modes with engineer approval.

04

Build and measure

Assemble a prototype and compare bench results with the original requirements.

CUSTOMER OUTPUT

A versioned design package + a measured demo

Measure elapsed development time, engineering hours and first-build test pass rate.

From a demo to a repeatable production release.

ROSE workflow 02 / From demo to mass production

01

Engineer for volume

Review tolerances, process capability, part reuse and assembly sequence.

02

Qualify the supply base

Link approved parts to quotes, lead times, origin records and alternates.

03

Build the control plan

Generate work instructions, fixtures and end-of-line tests from the released design.

04

Pilot, then release

Run a controlled lot. Approve yield, reliability and cost before increasing volume.

CUSTOMER OUTPUT

A manufacturing package with a traceable release record

Measure first-pass yield, cost per good unit, change-cycle time and field returns.

The U.S. factory scales by proven process.

Start with flexible production. Automate the stable, repetitive work as demand and process data justify investment.

01

Manufacturing partners

Qualify machining and assembly capacity. Validate fixtures, work instructions and origin records.

Gate: repeatable pilot lots

02

Dedicated U.S. cells

Bring critical assembly, calibration and test under direct process control.

Gate: demand + yield + unit economics

03

Lights-out processes

Automate selected stable operations with monitoring, exception handling and recovery.

Gate: sustained unattended reliability

ROSE links the released design to the actual factory process and its performance data.

Every production cycle improves the next.

PHASE 3 / LONG-TERM VISION

RSI

Recursive improvement of robot hardware and its manufacturing process

ROSE uses design, test and production outcomes to propose the next controlled revision.

Observe	Yield, torque, thermal behavior, failures and field returns
Propose	Design alternatives, process settings and approved part substitutions
Validate	Simulation, bench testing and controlled pilot lots
Release	Engineer approval, versioning, traceability and rollback

Success means lower cost per good unit and faster validated changes, with safety and quality held constant.

Three stages, with measurable release gates.

STAGE	BUILD	ADVANCE WHEN
01 **Now–2027** Hardware supply	Module evaluation kits U.S. manufacturing transfer Documented sourcing	Paid pilots convert to repeat orders. Released SKUs pass cost, quality and origin review.
02 **2028–2029** ROSE + U.S. cells	Two commercial software workflows Dedicated production capacity	Customer projects complete both workflows. Cell economics support additional automation.
03 **2030 onward** Adaptive manufacturing	Broader lights-out processes Validated improvement loops	Successive released revisions improve unit cost and yield with controlled reliability.

Planning sequence. Timing depends on product readiness, demand, financing and regulatory requirements.

Capital converts engineering into U.S. supply.

WORKING SEED FINANCING PLAN

$8–10M

Existing September 2026 financing range.
Final round terms remain subject to confirmation.

Fund the first repeatable U.S. production program and validate ROSE with real engineering work.

$10M PLANNING CASE

Factory and production equipment	$2M
Team and operating runway	$3M
R&D, materials and production transfer	$3M
Working-capital / contingency reserve	$2M

Capital deployment follows pilot demand and production gates. At the lower end, stage equipment purchases and reserve use.

Next evidence: paid pilots, repeat orders, qualified component records and measured production yield.

Internal: fundraising.json, updated 5 September 2026. Allocation is a planning case, not approved spend or a funded budget.

The hardware foundation for robot builders.

Manufacturing experience today.

ROSE-enabled engineering next.

A U.S. factory that improves with every release.

FlatHill jysong47@gmail.com

U.S. hardware supply platform

The 80% model has an explicit dependency.

QUADRUPED DESIGN SCENARIO

80 / 100 = **80%**

Sum the cost of qualifying domestic direct components. Divide by the cost of all direct components.

Use one component boundary. Avoid counting both an assembly and its internal parts. Group totals in this deck are presentation rollups.

80% 5 percentage points above the 75% comparison line.

01 **Same denominator**

Exclude final robot assembly cost, sales margin and corporate R&D from the component-cost denominator.

02 **Evidence per component**

Record manufacturer, manufacturing location, cost basis and origin support. Treat unknown origin as foreign.

03 **A supplier address is insufficient**

The 42-point joint contribution is conditional. If it fails, the illustrated domestic share falls to 38%.

FAR 25.003 and 25.101. Illustrative non-iron/steel-dominant, non-exempt cost scenario. No COTS waiver assumed.

Commodity inputs are an engineering choice.

META / MUSE GADGETS

An official ESP32 example

Meta’s open-source Muse Gadget SDK supports off-the-shelf ESP32 boards and Linux devices.

It demonstrates an available development ecosystem. It does not establish FlatHill robot authorization or future component eligibility.

Official repository ↗

TESLA / GLOBAL SUPPLY

Global sourcing can coexist with U.S. manufacturing

Tesla’s 2025 annual report describes parts from thousands of global suppliers. Its 2019 filing explicitly discussed U.S. production exposure to duties on Chinese components.

These are sourcing examples, not a robotics compliance precedent or a current China-content estimate.

2025 annual report ↗

Qualify each MCU, power device and motor input against the rules and product configuration in force at release.

Meta Muse Gadget SDK · Tesla 2025 10-K · Tesla 2019 Q2 10-Q.

Evidence and assumptions stay distinct.

Source-supported materials

Internal product drawings, module concepts, team profiles and prior financing plan. Official FCC notices, FAR provisions, Meta repository and Tesla filings.

Management representations

Module IP ownership, current team participation, FlatHill financing / promotion history and the early ROSE prototype.

Planning assumptions

All four BOM allocations, the 80% domestic target, roadmap dates, manufacturing sequence and future software revenue model.

Product release documentation

Obtain supplier origin records and a costed BOM. Verify component boundaries, U.S. manufacture and applicable authorization.

IP diligence

Review assignments, third-party licenses, U.S. filing records and counsel’s product-specific FTO opinion. Patent registration alone does not establish FTO.

U.S. origin marketing

The FTC’s unqualified “Made in USA” standard requires all or virtually all U.S. content. This deck’s 80% model does not establish that claim.

Full source notes and the editable planning data accompany this presentation. Status snapshot: 12 October 2026.

USPTO: patent rights · FTC: Made in USA Rule · Full source notes.